

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I S.Blessika(192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _S.Blessika

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Introduction

Cloud computing provides scalable resources that help businesses handle sudden increases in user traffic without affecting application performance.

Cloud Auto-Scaling

- Automatically increases or decreases the number of servers based on traffic demand.
- Ensures sufficient resources during peak shopping periods.
- Reduces costs by removing unused resources during low traffic.

Load Balancing

- Distributes incoming user requests evenly across multiple servers.
- Prevents server overload.
- Improves application availability and reliability.

Reducing Downtime and Improving User Experience

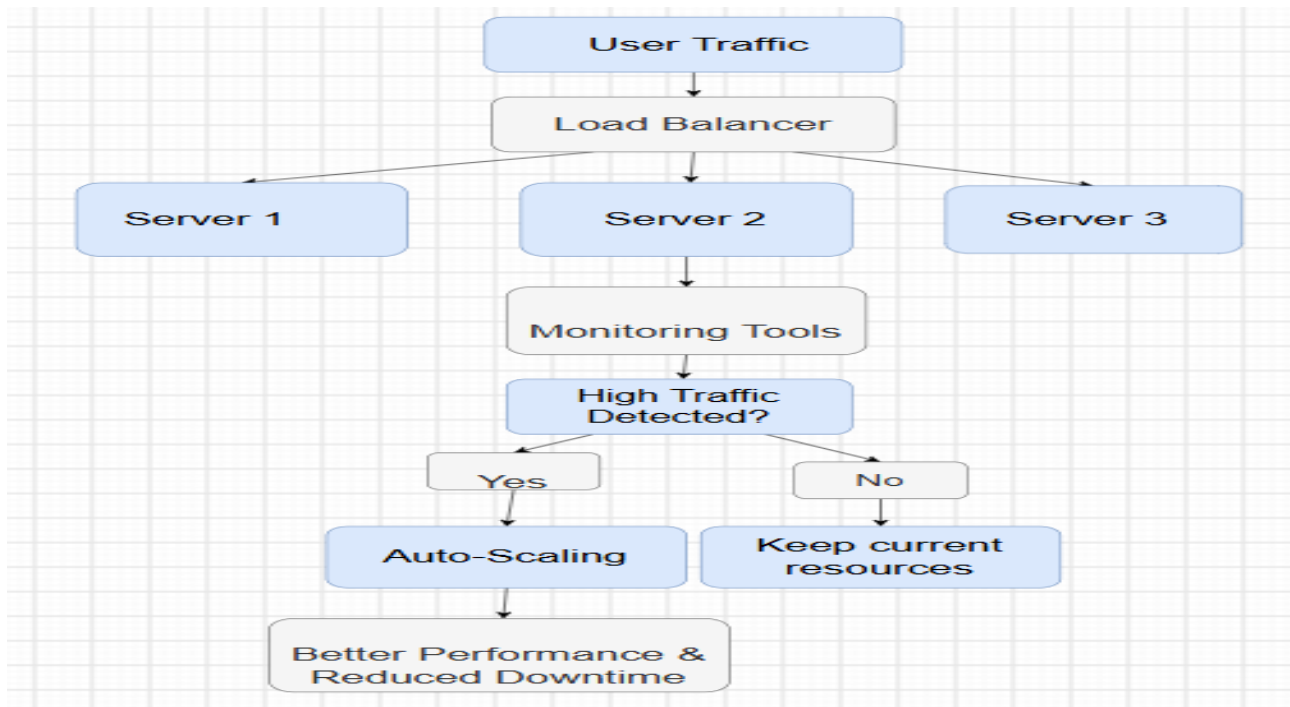
- Redirects traffic to healthy servers if one server fails.
- Prevents website crashes during heavy traffic.
- Provides faster page loading and smooth checkout experiences.

Role of Monitoring Tools

- Continuously monitor CPU usage, memory, response time, and network traffic.
- Trigger auto-scaling when resource usage exceeds predefined thresholds.
- Send alerts to administrators about system issues.

Real-World E-commerce Scenario

During **Black Friday** or **Diwali sales**, platforms like Amazon or Flipkart receive millions of visitors. Auto-scaling adds new servers automatically, while load balancing distributes customer requests efficiently, ensuring uninterrupted shopping.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Introduction

Cloud storage offers scalable, secure, and reliable solutions for storing large volumes of backup data while reducing infrastructure costs.

Durability and Data Redundancy

- Stores multiple copies of data across different servers and data centers.
- Protects data from hardware failures.
- Ensures high durability and availability.

Encryption and Access Control

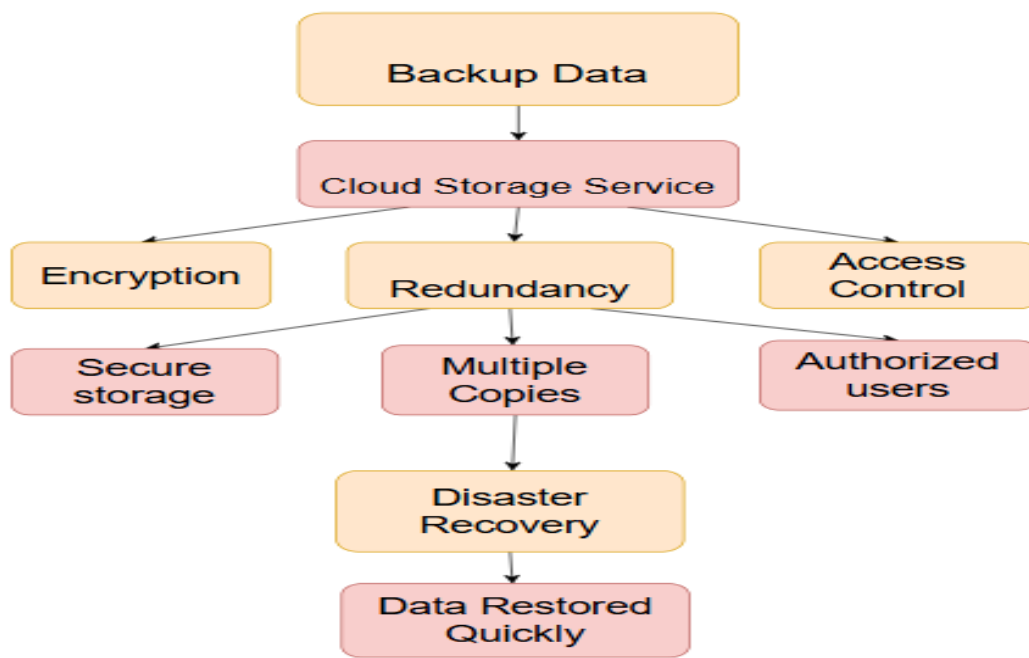
- Encrypts data while it is stored (at rest) and during transmission (in transit).
- Uses Identity and Access Management (IAM), role-based access control, and Multi-Factor Authentication (MFA).
- Prevents unauthorized access to sensitive backup data.

Cost Advantages

- Pay-as-you-go pricing model.
- No need to purchase expensive storage hardware.
- Lower maintenance and electricity costs.
- Easy scalability without additional infrastructure investment.

Disaster Recovery Example

A company stores daily database backups in the cloud. If its local data center is damaged by a flood or fire, it restores all critical data from the cloud and resumes business operations quickly.



3. A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction

Global applications require fast content delivery to users located in different regions. CDNs help achieve this efficiently.

Content Delivery Networks (CDNs)

- Store copies of application content in multiple geographic locations.
- Deliver content from the nearest server to the user.
- Reduce network congestion and latency.

Role of Edge Servers

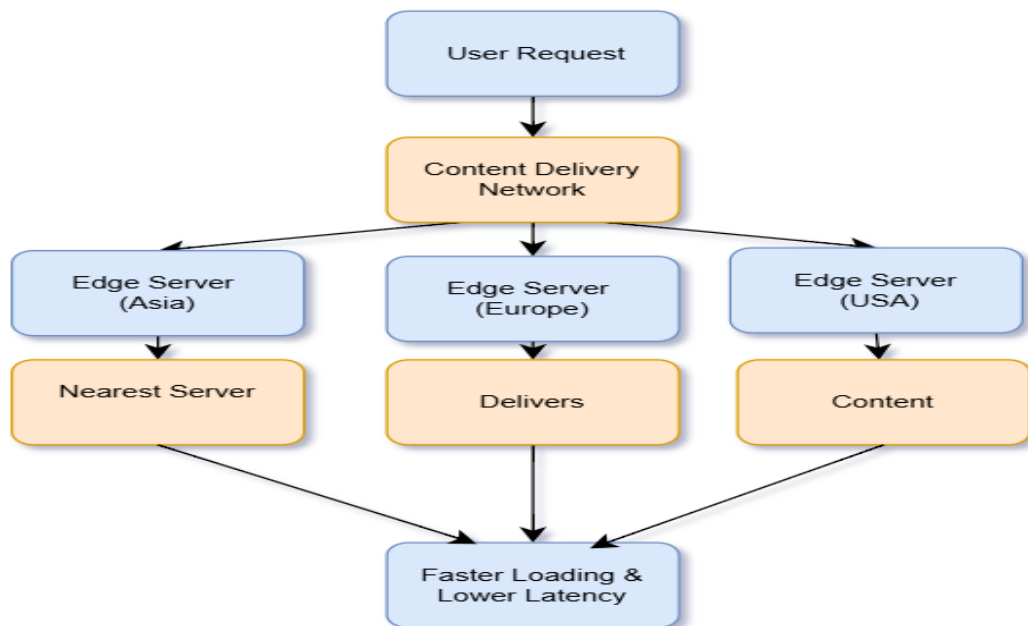
- Located close to end users.
- Minimize response time.
- Improve website and application loading speed.
- Reduce buffering and delays.

Global Availability

- Cloud providers maintain multiple regions and availability zones.
- Traffic can be redirected if one region becomes unavailable.
- Ensures continuous service worldwide.

Real-Time Streaming or Gaming Example

Netflix uses CDNs to stream videos from nearby servers, reducing buffering. Online multiplayer games use edge servers to reduce latency, providing smoother gameplay.



4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Financial institutions must protect customer information while complying with industry regulations and legal requirements.

Regulatory Compliance and Auditing

- Cloud providers support standards such as ISO 27001, PCI DSS, SOC 2, and GDPR.
- Regular audits verify compliance with security policies.
- Compliance reports help organizations meet regulatory requirements.

Role of Logging

- Records user activities, system changes, and financial transactions.
- Provides audit trails for investigations.
- Helps identify unauthorized access.

Role of Monitoring

- Detects suspicious activities and security threats.
- Sends real-time alerts.
- Supports quick incident response.

Role of Encryption

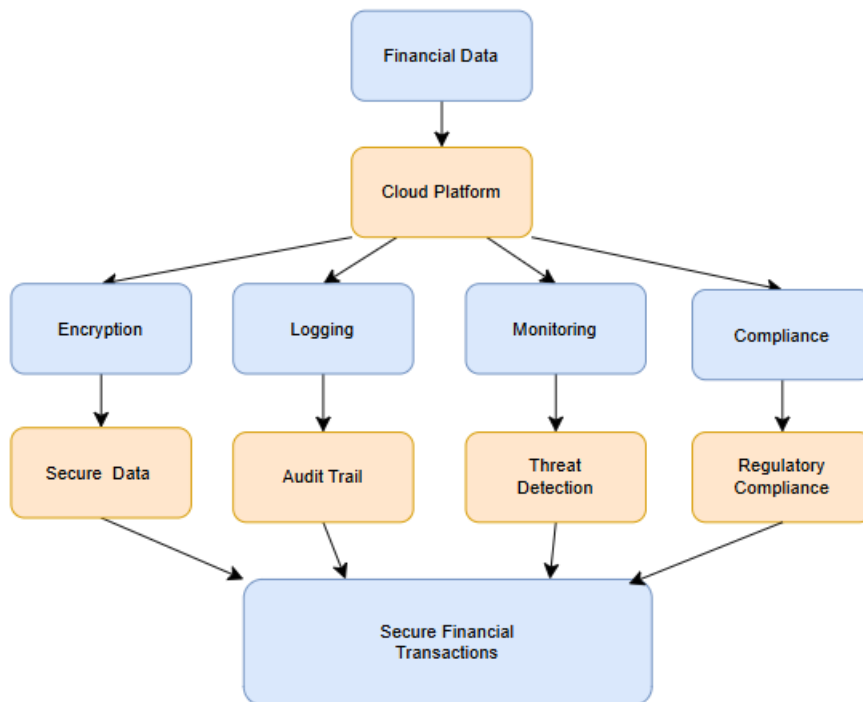
- Protects sensitive financial data during storage and transmission.
- Prevents unauthorized users from reading confidential information.

Data Sovereignty

- Stores customer data in approved geographic regions.
- Uses region-specific cloud data centers.
- Ensures compliance with local data protection laws.

Example

An online banking system encrypts customer account details, records every financial transaction, continuously monitors login attempts, and stores customer data within the required country to satisfy regulatory requirements.



5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Containerization packages applications and their dependencies into lightweight containers, making deployment faster and more efficient.

Containers vs. Virtual Machines

Containers	Virtual Machines
Share the host operating system	Include a complete operating system
Lightweight	Heavier and larger
Start in seconds	Slower startup
Use less CPU and memory	Consume more resources
High application density	Lower application density

Role of Kubernetes

- Automates container deployment.

- Performs auto-scaling based on demand.
- Restarts failed containers automatically.
- Supports load balancing.
- Enables rolling updates with minimal downtime.

Portability Across Clouds

- Containers run consistently on different cloud platforms.
- Eliminate environment compatibility issues.
- Support hybrid and multi-cloud deployments.

Microservices Example

An online shopping application consists of separate containerized services:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Management Service
- Notification Service

Each service can be independently developed, updated, and scaled according to demand.

